# "Tell Me What That Is": Examining Blind and Low Vision Individuals' Reviews on Assistive Smartphone Applications

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## Abstract

Smartphone applications (apps) designed for Blind and Low Vision (BLV) individuals play a crucial role in supporting their independence and accessibility. In this study, we adopted a mixed-methods approach, combining topic modeling, keyword-based filtering, and thematic analysis to analyze user-written reviews from 27 assistive smartphone apps focused on navigation and object/text recognition. We curated a list of 133 keywords related to visual impairment and eye conditions and extracted 2,163 reviews (out of 12,229 reviews). We analyzed 3,163 reviews (containing the extracted 2,163 reviews and a random 1,000 reviews that did not contain BLV-related keywords). Our findings reveal six major categories of user feedback, including praise and usefulness, general app usage, usability issues, feature requests, bug reports, and privacy concerns. From the 2,163 keyword-matched reviews, we manually annotated and isolated 680 reviews written by or about BLV individuals, revealing references to 20 distinct eye conditions. Our analysis shows that reviews involving BLV individuals appear more frequently in assistive apps than in general-purpose ones. While the overall sentiment of these 680 reviews was positive, BLV-authored reviews contained a higher proportion of critical feedback—particularly regarding usability—compared to reviews from non-BLV users, highlighting the need to prioritize their perspectives in app design.

## CCS Concepts

• **Human-centered computing → Empirical studies in accessibility**.

## Keywords

accessibility, blind, low vision, visual impairment, apps, user review, BLV

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## 1 Introduction

Smartphones have become an integral part of our lives. For Blind and Low Vision (BLV) individuals, smartphones can serve as an assistive tool and enhance accessibility and independence [33]. iOS and Android, the two major smartphone platforms, provide integrated assistive technology [19, 26, 33]. However, many applications (apps) do not provide adequate accessibility features [3, 21, 29, 33]. These shortcomings can often be identified through user-written reviews [8, 22]. Incorporating user feedback from the reviews has been shown to have a positive impact on app success [23].

Prior research has revealed how BLV individuals use smartphones [5, 25]. To explore accessibility challenges, researchers have analyzed Android app reviews related to accessibility [10] both manually [3] and with automated techniques [1, 2]. Although prior research has analyzed top Android app reviews to uncover insights related to accessibility [10, 21], these studies primarily focused on accessibility related reviews. As a result, there remains a gap in understanding the broader content of reviews for apps specifically designed for BLV individuals. Thus, in this work, we focus on two types of apps that BLV individuals can use in their regular life to enhance accessibility and independence: 1) apps for navigating and 2) apps for object/text recognition. We selected 27 such apps and analyzed their reviews to answer the following research questions:

**RQ1:** What do users express in reviews of assistive apps designed to help BLV individuals navigate and recognize objects? How can the reviews be categorized?

**RQ2:** Of all the reviews, how many involve or are written by BLV individuals? Is the presence of such reviews higher in this type of app? What type of eye conditions do they mention? What sentiment do the reviews express?

To address these research questions, we analyzed user reviews from 27 assistive smartphone apps developed to aid BLV individuals navigate or detect objects and text. We adopted a mixed methods approach that combined keyword-based filtering, topic modeling [6, 16], and thematic analysis [9]. Our findings reveal not only the diverse experiences of BLV individuals but also highlight the crucial role of BLV user feedback in improving assistive technologies.

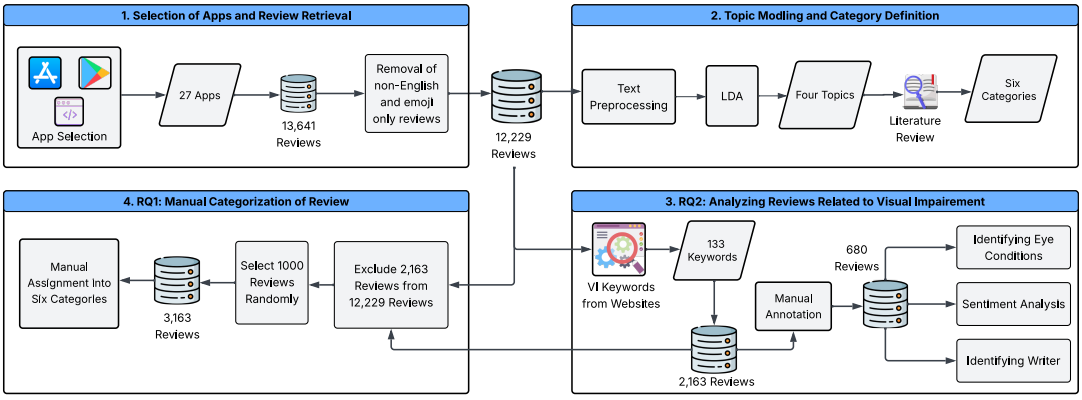



**Figure 1: Overview of the methodological workflow: (1) selection of 27 assistive apps and retrieval of 12,229 user reviews; (2) preprocessing and topic modeling to define review categories; (3) keyword-based filtering to identify 2,163 reviews related to visual impairment, including 680 written by or about Blind and Low Vision (BLV) individuals; and (4) top-down thematic analysis of 3,163 reviews to uncover user perspectives. Each step corresponds to a subsection in the Methodology.**

## 2 Methodology

Our methodology involved four key stages (Fig. 1): (1) identifying and selecting relevant apps from major app platforms and online sources; (2) retrieving, cleaning, and preparing user review data for analysis and creating categories; (3) identifying and analyzing reviews specifically referencing BLV individuals; and (4) applying topic modeling to assign categories to the reviews.

### 2.1 Selection of Apps and Review Retrieval

We focused on two categories of apps that BLV individuals can utilize to perform essential daily tasks: 1) navigation and 2) object and text recognition. We searched for the apps of the desired categories on Google Play [14] and Apple App Store [4]. Moreover, we explored online resources discussing apps helpful for BLV individuals [12, 17, 31]. We merged the results and inspected the descriptions of these apps on Google Play and the Apple App Store to ensure that they had been created exclusively for BLV individuals. Finally, we selected 27 apps (Table 3), among which ten were only available for iOS and two were only available for Android. Among the 27 apps, we included an app called *Be My Eyes* [11], which enables users to connect with sighted volunteers for assistance. Although the app is officially categorized under *Lifestyle* in the Apple App Store, it can be utilized for both navigation and object recognition.

We utilized two Python libraries [20, 30] to extract publicly available app reviews from Google Play and the Apple App Store, resulting in 13,641 reviews (7,562 from Google Play and 6,079 from the Apple App Store) written between 2010 and 2025. Five of the selected apps did not have any reviews. Non-English and emoji-only reviews were excluded, resulting in 12,229 reviews.

The app reviews were publicly available. Thus, IRB approval was not required. To ensure anonymity, we only stored the reviews, user ratings, and review dates, excluding any personally identifiable information (e.g., username, profile picture). Moreover, the reviews quoted in this paper's findings were paraphrased in such a way that preserves the original meaning but prevents the identification of the original authors.

### 2.2 Topic modeling and Review Category Creation

We adopted a topic-modeling approach to categorize the 12,229 reviews using the Latent Dirichlet Allocation (LDA) model [6, 16]. We pre-processed the reviews in three steps — tokenization, lemmatization, and stop word elimination [15, 24] and then ran the model [27] on the processed data. We used a coherence score (CV metric) [28] to evaluate the results. To achieve a good coherence score [7], we ran the model ten times and fine-tuned it. Moreover, from the initial results, we identified keywords — *app, good, great, application, use, using, love, like, thank, thanks, work* — that appeared frequently in the topics and made them less interpretable. We reran the whole process by adding these with the stop words and were able to improve the coherence score. We ran the LDA model separately for the reviews of the two app categories (*Be My Eyes* was considered with navigation). Finally, we retrieved six topics for the navigation app reviews (coherence=0.56) and five topics for the object and text recognition app reviews (coherence=0.61). From the keywords for each topic, we determined a category (Table 1 and 2). Merging these two, we developed four categories: **(1) Praise and usefulness** - reviews expressing satisfaction with the app, or describing how it is helpful, or praising it; **(2) General app usage** - reviews that describe everyday usage or app functionality; **(3) Usability issues and negative feedback** - reviews that describe the challenges in using the app, or features and styles that the users do not like; **(4) Feature requests** - reviews suggesting new features or asking for improvements to existing functionality. Additionally, we included two more categories from prior research [18, 21, 22]: **(5) Bug reports** - where users report app crashes, unexpected behaviors, etc., and **(6) Privacy and security issues** - reviews that mention data privacy, permissions or trust/security concerns.

### 2.3 Identifying Reviews Related to BLV Individuals

We identified reviews that involved users with visual impairments or eye-related conditions. We compiled a list of eye conditions using

the "Glossary of Eye Conditions" from the American Foundation for the Blind [13] and "Common Vocabulary in Eye Reports" by [32]. We considered all the terms related to the eye and diseases of the eye, yielding 133 keywords. We conducted a keyword search across all reviews and found 2,163 reviews (out of 12,229) that had at least one keyword from the list and manually inspected these reviews. Reviews with vague references, such as those stating that the app was helpful to blind individuals without further context, were excluded. In these cases, the reviewer's identity was unclear, and it was uncertain whether they had first-hand experience or direct involvement with a BLV user who benefits from the app. There were 680 reviews involving BLV individuals. Each of these reviews was labeled with the sentiment they expressed (positive, negative, and neutral), the eye condition (e.g., legally blind, macular degeneration, etc.) mentioned in it, and the reviewer (BLV individual, their family members or friends, or someone else).

## 2.4 Categorization of App Reviews

We filtered the reviews (2,163) related to visual impairment from all the reviews (12,229). From the remaining reviews that did not have any keywords related to visual impairment, we randomly selected 1,000 reviews, each containing at least five words. These 1,000 random reviews were then merged with the 2,163 reviews related to visual impairment, yielding a data set of 3,163 reviews. These reviews were analyzed manually in a deductive (top-down thematic analysis) approach [9]. The categories that we created were used as themes.

# 3 Results and Discussion

## 3.1 RQ1: Categories of the App Reviews

The reviews (n=3,163) were assigned one or more categories (Fig. 2a). Some reviews (n=100) were irrelevant, out of context, and insubstantial (e.g., *"how do blind people install apps?"*); thus, they were marked as outliers. The insights from the analysis of the rest os the reviews are presented in the following subsubsections:

*3.1.1 Praise and usefulness (n=2142).* The majority of the reviews fall in this category. In these reviews, the users have expressed satisfaction and love for the app. These often included positive feedback emphasizing usefulness and accessibility. In some of the reviews, gratitude to the developers was expressed: *"I really appreciate this tool; it's the most helpful one for people like me with dyslexia."* (user review, *Seeing AI*). Another user wrote, *"I think this app could dramatically improve accessibility for blind people everywhere."* (user review, *Lookout - Assisted Vision*). Users also mentioned how the apps enabled them to gain independence, which they value greatly: *"Blind users often hesitate to seek assistance, but this app enables us to be self-reliant."* (user review, *TapTapSee*). A significant presence of this category suggests that these apps have a positive impact on the everyday activities of BLV individuals, enhancing their accessibility and independence. These reviews also reveal user-favored features that developers can leverage to enhance their apps.

*3.1.2 General app usage (n=558).* This category includes reviews about how the app is used or what can be done using the app. Users often shared specific scenarios, detailing the tasks they completed using the app: *"I use this app daily during my commute from home to college to read signs at train stations, and also in class to help me find the lecture halls."* (user review, *Lookout - Assisted Vision*). Another user mentioned, *"I assisted a woman by reading her prescription labels so she could identify the correct medication."* (user review, *Be My Eyes*). The reviews in this category indicate that users rely on the apps for a variety of everyday tasks, such as outdoor and indoor navigation, identifying groceries, clothes, and more. Future research could provide detailed insights into how BLV individuals utilize these apps in outdoor and indoor environments.

*3.1.3 Usability issues and negative feedback (n=489).* In these reviews, users complained about features they disliked, such as the user interface (UI) (e.g., no dark theme, small fonts, low contrast). Some mentioned apps being less accessible: *"Tabular and Excel data should be presented in a more accessible way."* (user review, *Envision AI*). Pricing was also a prevailing concern, with BLV individuals mentioning that they already have many accessibility-related expenses, making it difficult to purchase additional apps. Users also complained about apps being laggy, incompatibility with older phones, video and image quality, etc. These reviews offer valuable insights for developers by highlighting less desired features by the users and opportunities to enhance the user experience. However, user experience varies significantly, and customizable accessibility features are desired: *"Not every blind person does things the same way. We should be able to choose the accessibility features that suit us."* (user review, *Be My Eyes*). Thus, the design process of these apps should actively involve BLV individuals.

*3.1.4 Feature requests (n=208).* Users requested a wide range of features, including support for additional languages (e.g., Arabic, Bengali, French, Indian regional languages), capability of identifying different currencies, integration with smart glasses, offline functionality, facial recognition from photos, image sharing and importing, alternative notification methods, enabling volunteers to set availability, support for reading PDF and DOC files, etc. Some apps are only available for iOS, and users have asked for an Android version. A significant presence of reviews with language support indicates that not all BLV individuals have equal opportunities, particularly those in underdeveloped regions. Future research should explore accessibility issues, technology usage, and the specific needs of BLV individuals from underdeveloped regions.

*3.1.5 Bug reports (n=89).* Users reported various unexpected behaviors of the apps, including glitches during sign-up or sign-in, unresponsive UI elements such as buttons, crashes on particular devices, notification issues, inaccuracy in image recognition, location inaccuracy, and problems introduced by buggy updates. Some reviews mentioned bugs without providing specific details. Bugs can be found in all types of software. However, bugs in assistive apps can be dangerous, as one user reported that the app provided them with the results of a past image scan where they were scanning medications: *"I tried to use it to read my prescription, but instead it showed me an old one that I scanned previously. It's really risky if it tells me I'm taking the wrong medicine."* (user review, *Be My Eyes*).

*3.1.6 Privacy and security issues (n=56).* All the reviews of this category belong to the app *Be My Eyes*. Users raised concerns about the app sharing information with third-party organizations. One user mentioned the possibility of confidential data (e.g., credit card

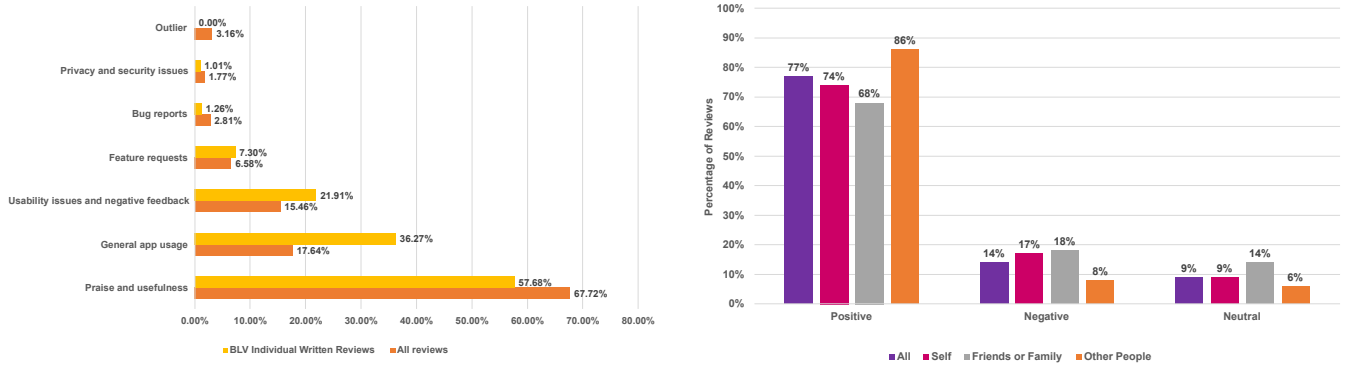



**Figure 2: a) Category distribution of reviews across all reviews (n=3,163) and the subset written by BLV individuals (n=397). While "Praise and Usefulness" dominates both groups, BLV-written reviews show higher proportions of "General App Usage" and "Usability Issues," offering deeper insights into real-world use cases and challenges faced by BLV users. b) Sentiment distribution of reviews involving BLV individuals, categorized by overall authors, BLV individuals themselves, their family and friends, and other people**

information) being exposed when a BLV individual seeks sighted help. One visually impaired user wrote that the app refused to read out credit card information to preserve privacy and suggested calling a volunteer: *"As a blind user, I need to check my credit card details, but the app refuses for privacy reasons and tells me to call a volunteer. Why share sensitive info with a stranger? An AI reading it would be more secure"*. The most common privacy issue was people who faked visual impairment and played a prank or sexually harassed volunteers, highlighting the necessity for user verification during the onboarding process in remote sighted assistance apps. The absence of privacy concerns in other app reviews calls for further research. Interviewing users of those apps may help surface privacy concerns.

## 3.2 RQ2: Insights from Reviews Involving BLV

Of the 2,163 reviews containing at least one keyword related to visual impairment or eye conditions, 680 reviews involved BLV users. These reviews were either written by BLV individuals (n=397), or their family and friends (n=65), or by another person (n=218). A total of 20 eye conditions (listed in A.1) were mentioned. These included, but were not limited to, total or legal blindness, color blindness, partial sight, having low vision, albinism, macular degeneration, retinal detachment, glaucoma, and retinitis pigmentosa.

Among all the reviews, 17% had at least one keyword related to visual impairment. 5.5% of the total reviews explicitly involved BLV individuals. In contrast, previous research on general-purpose apps found that less than 0.1% of the reviews included keywords related to visual impairment [21], suggesting that assistive apps have more reviews involving BLV individuals.

The sentiment analysis of reviews related to BLV individuals showed an overall positive impression (Fig. 2b), even when reviews were grouped by author, suggesting the usefulness of the apps. However, we observed a significantly higher proportion of negative reviews among those written by BLV individuals or their family and friends, compared to those written by others.

We specifically examined the perspective of BLV individuals by comparing the distribution of review categories across all reviews and those written by BLV individuals (Fig. 2a). Most of the reviews fall into the first category, where users expressed how useful the app is. The proportions across categories are generally similar, except for the second and third categories, which are more prominent in BLV-authored reviews. BLV individuals shared a lot about how they use these apps, contributing to the higher prevalence of the second category. These reveal real-world usage contexts that can help app developers improve app functionality and guide Software Quality Assurance (SQA) teams in designing realistic test cases. The third category also has a significant presence, with expressed usability issues. This may explain why the number of negative reviews written by BLV individuals is higher compared to other people. However, only 11 (0.08% of the total) BLV-authored reviews were related to accessibility, which aligns with previous research [10, 21].

## 4 Conclusion

We presented an analysis of user-written reviews from 27 assistive smartphone apps designed for BLV individuals. We categorized over 3,000 reviews and identified six prominent themes. Our findings reveal that a substantial portion of the reviews—especially those authored by BLV individuals or their close networks—highlight both the value and the limitations of these apps. We found that the majority of reviews expressed positive sentiment, particularly about the apps' usefulness in fostering independence and accessibility. In contrast to prior studies that examined general-purpose apps, our approach showed a significantly higher prevalence of BLV-related content and user engagement in these niche applications. Future research could expand the scope to include additional app categories beyond navigation and object recognition, incorporate other review platforms and languages, and involve collaborative annotation to minimize individual bias. Despite limitations related to app selection imbalance (72% of the reviews referred to one app, five of the apps did not have reviews, and 11 apps had fewer than

20 reviews), our findings highlight the importance of centering user feedback—especially from marginalized communities—in the development and refinement of assistive digital tools.

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# A Appendix

## A.1 Eye Conditions

We found 20 distinct eye conditions mentioned in the reviews — *vision, partially sighted, low vision, impaired, blind, blindness, legally blind, totally blind, glaucoma, macular degeneration, peripheral vision, impairement, VI, retinal detachment, visual acuity, retinitis pigmentosa, glare, albinism, color blindness, and CVI*. Moreover, tools such as *braille* and *lens* were mentioned. *Dyslexia*, a neurological condition, was mentioned, which is not a vision disorder but presents challenges similar to those experienced by BLV individuals.

## A.2 Topic Modeling Results

## A.3 List of selected apps

**Table 1: Navigation app review categories and associated keywords. The six topics were identified through Latent Dirichlet Allocation (LDA) modeling applied to navigation app reviews.**

| Category | Keywords |
|---|---|
| The review expresses that the app is helpful for blind people | blind, helpful, useful, eye, person, people, microsoft, life, make, developer |
| The users describing challenges related to some feature | help, call, get, answer, time, phone, receive, month, need, try |
| The review is about adding new features to the app | feature, time, add, ai, option, language, star, update, want, find |
| The review praises the app | help, people, need, able, amazing, way, volunteer, easy, feel, idea |
| The review is about content recognition | read, tell, product, mail, identify, service, try, person, ask, email |
| The review is about volunteering (a functionality of an app) | friend, get, know, today, volunteer, think, nice, recommend, help, hear |

**Table 2: Object and text recognition app review categories and associated keywords. The five topics were derived from LDA modeling of reviews for object and text recognition apps.**

| Category | Keywords |
|---|---|
| The user is praising the app's features | blind, people, picture, thing, amazing, tell, think, see, money, try |
| The review is about the app's functionality | picture, read, take, pay, time, scan, well, get, free, thing |
| The review is about adding new features to the app; The review is about the usefulness of the app | blind, read, help, identify, picture, add, language, helpful, useful, easy |
| The review discusses general app usage or usability challenges | read, thing, picture, time, text, know, find, want, blind, file |
| The review is about document scanning or text reading improvements | time, read, update, document, scan, text, page, try, buy, fix |

**Table 3: List of assistive apps analyzed in this study. Each entry includes the app name, platform availability (iOS, Android, or both), and its primary function (navigation or object/text recognition). The selection process ensured that all apps were designed specifically to support BLV individuals in their daily activities.**

| Name | Category | Platform | Total Reviews Collected | Total Reviews Categorized | Total Reviews Related BVI Individuals |
|---|---|---|---|---|---|
| Lazarillo Accessible GPS | Navigation | Android, iOS | 133 | 50 | 17 |
| Blind Explorer | Navigation | Android, iOS | 12 | 3 | 0 |
| Be My Eyes | Lifestyle | Android, iOS | 8820 | 2125 | 351 |
| Vision: for blind people | Navigation | Android, iOS | 14 | 4 | 0 |
| BlindSquare | Navigation | iOS | 83 | 39 | 17 |
| Loadstone GPS | Navigation | iOS | 2 | 0 | 0 |
| VoiceVista | Navigation | iOS | 15 | 2 | 1 |
| Oko - Cross streets and Maps | Navigation | iOS | 82 | 28 | 17 |
| NaviBlind | Navigation | iOS | 0 | 0 | 0 |
| RightHear - Blind Assistant | Navigation | Android, iOS | 32 | 9 | 1 |
| WeWALK - Smart Map | Navigation | Android, iOS | 31 | 7 | 2 |
| Seeing Eye GPS | Navigation | iOS | 37 | 15 | 6 |
| Seeing Assistant Move | Navigation | Android, iOS | 3 | 2 | 1 |
| All_Aboard | Navigation | Android, iOS | 0 | 0 | 0 |
| AI Guide Dog | Navigation | iOS | 0 | 0 | 0 |
| NOA Companion | Navigation | Android, iOS | 0 | 0 | 0 |
| OpenWay | Navigation | Android | 0 | 0 | 0 |
| Seeing AI | Object and text recognition | Android, iOS | 631 | 225 | 85 |
| Lookout - Assisted vision | Object and text recognition | Android | 469 | 125 | 24 |
| NaviLens | Object and text recognition | Android, iOS | 39 | 12 | 3 |
| Envision AI | Object and text recognition | Android, iOS | 623 | 161 | 35 |
| Supersense - AI for Blind | Object and text recognition | iOS | 107 | 30 | 8 |
| OneStep Reader | Object and text recognition | Android, iOS | 361 | 93 | 30 |
| TapTapSee | Object and text recognition | Android, iOS | 653 | 206 | 73 |
| Seeing Assistant Home | Object and text recognition | Android, iOS | 11 | 3 | 1 |
| Digit-Eyes | Object and text recognition | iOS | 29 | 8 | 5 |
| Cash Reader for the Blind | Object and text recognition | iOS | 42 | 16 | 6 |